

The empirical recurrence for OEIS A283413, recovered from its tail

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8 September 2026

Abstract

OEIS A283413 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 258$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 78$. Its last coefficient is $c_{72} = -746496 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A283413 is “Number of $n \times 6$ 0..1 arrays with no 1 equal to more than three of its horizontal, vertical and antidiagonal neighbors.”. It has offset 1 and begins

64, 3525, 146228, 6865368, 315717408, 14511646624, 668024193788, 30738576483000, ...

The entry states:

Empirical recurrence of order 72 (see link above).

As of the “Last modified” line on the live entry (May 24 2024 15:57:49, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 258$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 258$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 516$ exact terms from $n = 6$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	21	c_{19}	52040557146246	c_{37}	356825801928812110	c_{55}	−1548602866967648
c_2	866	c_{20}	−2456386807457452	c_{38}	−888852677067783950	c_{56}	−333420271495360
c_3	11365	c_{21}	12076874109375335	c_{39}	1260463533659070547	c_{57}	572369064456832
c_4	74118	c_{22}	−37813662453495726	c_{40}	−1078903502932144509	c_{58}	−197806498998336
c_5	303343	c_{23}	91916976155526079	c_{41}	−167343812883184982	c_{59}	92983295362752
c_6	406103	c_{24}	−181648904684728585	c_{42}	816927779418668011	c_{60}	−28581980682624
c_7	−171302	c_{25}	300204261723421666	c_{43}	−207735103973022156	c_{61}	13174699958160
c_8	−10184352	c_{26}	−435341830199211427	c_{44}	−254989658371545340	c_{62}	−3846742296768
c_9	76524805	c_{27}	594988823965303642	c_{45}	519379278757556088	c_{63}	885229255296
c_{10}	−569253230	c_{28}	−813977718412686718	c_{46}	−422809250249341808	c_{64}	260904059136
c_{11}	1988164150	c_{29}	1006734824414088977	c_{47}	−51958672581925720	c_{65}	59486402304
c_{12}	−41905210991	c_{30}	−953379872290638063	c_{48}	176666066346316160	c_{66}	−26657126784
c_{13}	202759667229	c_{31}	718733559015384651	c_{49}	−98140001798174224	c_{67}	1706323968
c_{14}	−507985354117	c_{32}	−583186381968250298	c_{50}	13543244337745144	c_{68}	−1442831616
c_{15}	−2596739810278	c_{33}	634864419571497493	c_{51}	58996254674864112	c_{69}	−146022912
c_{16}	22878880665419	c_{34}	−685657537256331010	c_{52}	−43811066648835936	c_{70}	−18931968
c_{17}	−94737667104768	c_{35}	328753778093773413	c_{53}	15183744426160608	c_{71}	13872384
c_{18}	198387908937694	c_{36}	9983470958131371	c_{54}	−2575882886910768	c_{72}	−746496

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 78$, and it is the recurrence OEIS A283413 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = 746496$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -746496 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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